Emma Fries, EIT

emmajeannefries@gmail.com (916) 756-7002 linkedin.com/in/emmafries734

Education

California Polytechnic State University - San Luis Obispo, CA June 2024
Bachelor of Science in Mechanical Engineering

Certifications

Engineer-in-Training (#186647) Dec 2025
Certified SolidWorks Associate (CSWA) - Mechanical Design Oct 2025

Skills

Design & Analysis: SolidWorks, AutoCAD, FEA (SolidWorks Simulation), GD&T, DesignBuilder, MATLAB, Python, Excel
Fabrication: 3D Printing (FDM), Prototyping, Manual Machining (Mill, Lathe), CNC Machining (Introductory)

Work Experience

Project Administrator Aug 2024 – Feb 2025
Rubicon Partners Inc. Sacramento, CA

- Assisted in due diligence and acquisition support for a large-scale high-rise commercial property, managing documentation, overseeing leases and contracts, and coordinating with tenants
- Collaborated with the project team to support financial and lease operations, including invoice processing, expense tracking, and maintaining accurate documentation
- Developed expertise in building's audio-visual system, becoming the go-to resource for tenants and leading tutorials to improve user proficiency and system efficiency
- Directed office operations, emails, and tenant relations with professionalism, ensuring prompt responses and effective resolution of inquiries

Projects

Visual Inspection System Senior Project - Edwards Lifesciences (NDA) Sept 2023 – June 2024

- Collaborated on a student team to design and prototype an automated vision system to detect manufacturing defects in heart valve components, delivering a proof-of-concept prototype demonstrating the feasibility of an automated inspection system for the engineering team
- Designed custom pneumatic fixtures, Meca500 robot motion control, and optical inspection assembly, ensuring precise integration of mechanical and robotic systems
- Modeled and designed components in SolidWorks and created GD&T technical drawings for fabrication via 3D printing and manual machining, enabling accurate and functional assembly
- Implemented robot motion and pneumatic control using RoboDK and Mecademic software, integrating Basler Pylon image acquisition to generate high-precision inspection images
- Presented final prototype and key findings to project managers, supporting potential adoption of automated inspection processes

Cross-Flow Water Turbine - California Polytechnic State University Apr 2024 – June 2024

- Built and tested a Banki-Michell water turbine system to meet competition requirements, successfully generating sufficient power to pull a 1 N load over 20 feet
- Designed the turbine, nozzle, and pulley system in SolidWorks, applying maximum power transfer principles and EES calculations while working within constraints on reservoir height, water volume, and component dimensions
- Fabricated the pulley and structural test stand using woodworking tools and prototyped key components (nozzle, turbine, reservoir) via 3D printing, enabling rapid iteration
- Conducted testing and implemented design improvements, resulting in increased turbine efficiency and faster pulling speed
- Produced a technical report detailing assembly procedures, design calculations, and testing results for the final turbine system